INSIGHTLENS: AI-POWERED VIDEO INTELLIGENCE AGENT WITH RETRIEVAL-AUGMENTED GENERATION (RAG)

ABSTRACT

In today's digital environment, organizations, educational institutions, and professionals generate large volumes of video-based content such as meetings, webinars, lectures, interviews, and discussions. Extracting meaningful information from these recordings is often time-consuming and inefficient. Manual review of lengthy recordings results in productivity loss and increases the possibility of overlooking critical information.

InsightLens is an AI-powered Video Intelligence Agent developed to automatically transform unstructured video and audio recordings into structured, searchable, and actionable knowledge. The system accepts YouTube video links or local media files as input and performs a complete intelligence pipeline including audio extraction, speech transcription, automatic title generation, meeting summarization, action item extraction, key decision identification, open question detection, and conversational question answering through Retrieval-Augmented Generation (RAG).

The solution integrates multiple state-of-the-art AI technologies including Whisper, Sarvam AI, Mistral Large Language Models, ChromaDB, HuggingFace Embeddings, LangChain, and Streamlit. Unlike traditional video summarization systems, InsightLens enables users to interact directly with meeting content through a conversational interface, making information retrieval significantly faster and more efficient.

The project demonstrates practical implementation of Natural Language Processing, Speech Recognition, Information Retrieval, Vector Databases, Large Language Models, and Retrieval-Augmented Generation within a unified intelligent platform.

1. INTRODUCTION

The rapid growth of online communication platforms has resulted in an exponential increase in recorded meetings, webinars, lectures, and video-based discussions. Although these recordings contain valuable information, extracting actionable insights often requires significant manual effort.

Professionals frequently spend considerable time revisiting recordings to identify important discussions, action items, and decisions. Traditional transcription systems provide raw text but fail to transform information into meaningful knowledge. Consequently, organizations require intelligent systems capable of understanding, analyzing, and retrieving information from recorded conversations.

InsightLens addresses this challenge by providing an end-to-end AI-powered video intelligence solution. The system not only transcribes spoken content but also generates summaries, identifies key discussion points, extracts actionable tasks, and allows users to ask natural language questions about the recording through an intelligent Retrieval-Augmented Generation pipeline.

By combining speech recognition, large language models, semantic retrieval, and conversational AI, InsightLens transforms passive recordings into interactive knowledge repositories.

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2. PROBLEM STATEMENT

Modern organizations encounter several challenges when managing recorded meetings and video content:

- Reviewing lengthy recordings requires significant time and effort.
- Important decisions are often difficult to locate within large transcripts.
- Action items and responsibilities may be overlooked.
- Follow-up questions remain undocumented.
- Traditional search mechanisms rely on keyword matching rather than semantic understanding.
- Users cannot interact naturally with meeting content.

Existing transcription tools primarily focus on speech-to-text conversion and lack advanced intelligence capabilities. There is therefore a need for an intelligent system capable of extracting actionable insights and supporting conversational access to meeting knowledge.

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3. OBJECTIVES

The primary objectives of the project are:

Functional Objectives

- Accept YouTube videos and local media files as input.
- Automatically extract audio content.
- Generate accurate speech transcripts.
- Produce concise meeting summaries.
- Generate professional meeting titles.
- Extract action items with assigned ownership.
- Identify important decisions made during discussions.
- Detect unresolved questions and follow-up topics.

- Enable conversational querying of meeting content.

Technical Objectives

- Develop a scalable AI processing pipeline.
- Integrate multiple speech transcription engines.
- Implement Retrieval-Augmented Generation.
- Build a semantic vector database for knowledge retrieval.
- Design an intuitive and user-friendly interface.
- Minimize hallucinations through context-grounded question answering.

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4. SYSTEM ARCHITECTURE

The architecture of InsightLens consists of multiple interconnected modules responsible for audio processing, speech recognition, language understanding, information extraction, and conversational retrieval.

Workflow:

Input Source (YouTube URL / Local File)
↓
Audio Extraction and Conversion
↓
Audio Chunking
↓
Speech Transcription (Whisper / Sarvam AI)
↓
Transcript Generation
↓
Title Generation
↓
Meeting Summarization
↓
Action Item Extraction
↓
Decision Extraction
↓
Open Question Identification
↓
Vector Database Construction
↓
Retrieval-Augmented Generation
↓
Conversational Question Answering
↓
Streamlit User Interface

The modular architecture ensures scalability, maintainability, and efficient processing of long-duration recordings.

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5. SYSTEM IMPLEMENTATION

5.1 Audio Processing Module

File: audio_extractor.py

The Audio Processing Module serves as the entry point of the system. It is responsible for obtaining audio data from external sources and preparing it for speech recognition.

The module supports two input formats:

1. YouTube URLs
2. Local audio/video files

For YouTube sources, the system uses yt-dlp to download high-quality audio streams. The downloaded content is automatically converted into WAV format using FFmpeg integration.

For local files, Pydub is used to convert the media into a standardized format consisting of:

1. Mono Channel Audio
2. 16 kHz Sampling Rate

This standardization ensures compatibility with speech recognition models.

To handle long recordings efficiently, audio files are segmented into ten-minute chunks. Chunking reduces memory consumption and improves processing scalability.

5.2 Speech Intelligence Module

File: speech_transcriber.py

The Speech Intelligence Module converts audio segments into textual transcripts.

The system incorporates two transcription engines:

Whisper Integration

For English recordings, OpenAI Whisper is utilized. Whisper provides high transcription accuracy and operates locally, ensuring reliable speech recognition.

Sarvam AI Integration

For Hinglish content, the system uses Sarvam AI's speech-to-text translation service. Since the Sarvam API only accepts audio segments shorter than thirty seconds, the system automatically divides audio chunks into twenty-five-second segments before transmission.

This design enables support for multilingual and code-mixed speech while maintaining transcription quality.

A dynamic routing mechanism automatically selects the appropriate transcription engine based on the language selected by the user.

5.3 Intelligent Summarization Module

File: meeting_summarizer.py

The summarization component employs Mistral Small Latest through LangChain to generate professional meeting summaries.

Due to context limitations of language models, transcripts are divided into smaller segments using Recursive Character Text Splitting.

The summarization process follows a Map-Reduce strategy:

Map Phase

Each transcript chunk is summarized independently.

Reduce Phase

The generated partial summaries are combined and synthesized into a final comprehensive meeting summary.

This approach enables processing of lengthy recordings while preserving contextual understanding.

The module also generates a concise professional meeting title limited to eight words.

5.4 Meeting Intelligence Analyzer

File: transcript_analyzer.py

This module transforms raw transcripts into actionable business intelligence.

Three specialized analysis pipelines are implemented:

Action Item Extraction

The system identifies:

1. Task descriptions
2. Responsible individuals
3. Deadlines when available

Key Decision Extraction

The system extracts important decisions, approvals, and strategic conclusions discussed during the meeting.

Open Question Detection

The module identifies unresolved concerns, pending discussions, and topics requiring future follow-up.

These outputs significantly improve post-meeting productivity and accountability.

5.5 Semantic Knowledge Base

File: retrieval_store.py

To support conversational querying, transcript data is transformed into vector embeddings and stored within a semantic database.

The transcript is divided into chunks of 500 characters with an overlap of 50 characters to preserve contextual continuity.

The system utilizes the HuggingFace embedding model:

all-MiniLM-L6-v2

Each chunk is converted into a high-dimensional semantic representation and stored in ChromaDB.

The vector database enables meaning-based retrieval rather than traditional keyword matching.

5.6 Retrieval-Augmented Generation (RAG)

File: rag_pipeline.py

The Retrieval-Augmented Generation pipeline provides conversational access to meeting knowledge.

When a user submits a question, the system performs the following operations:

1. Convert the query into an embedding.
2. Retrieve the most semantically relevant transcript segments.
3. Inject retrieved context into a prompt template.
4. Generate a response using the Mistral language model.

To reduce hallucinations, the model is explicitly instructed to answer only from retrieved transcript content. If the requested information is unavailable, the system informs the user accordingly.

This architecture ensures accurate and context-grounded responses.

5.7 Pipeline Orchestration

File: main.py

The Main Pipeline coordinates all processing stages.

The workflow includes:

- Audio Processing
- Speech Transcription
- Title Generation
- Summary Generation
- Action Item Extraction
- Decision Analysis
- Question Identification
- Vector Store Creation
- RAG Initialization

This centralized orchestration ensures smooth execution of the complete intelligence pipeline.

5.8 User Interface

File: app.py

The user interface is developed using Streamlit.

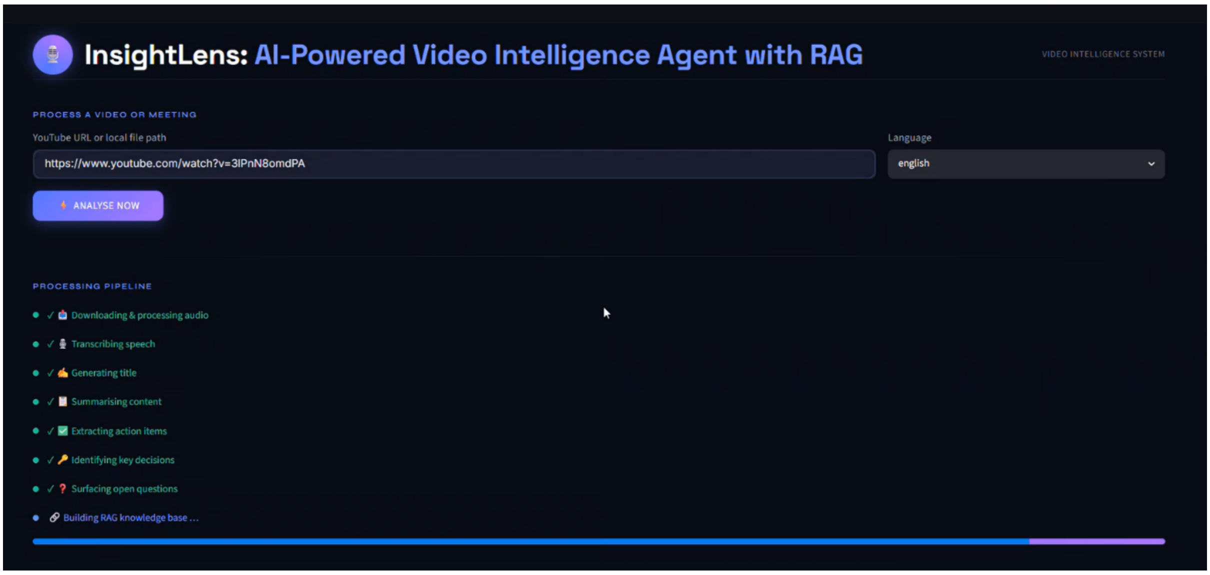
The application provides a modern dashboard featuring:

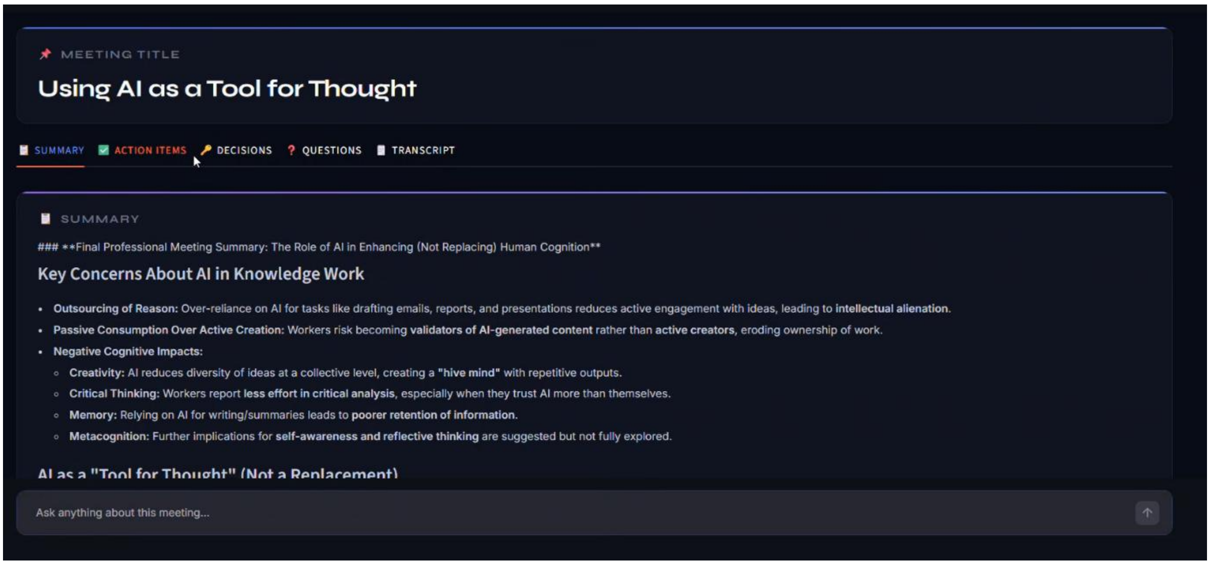
- Dark-themed professional design
- Real-time pipeline progress visualization
- Interactive result tabs
- Transcript download functionality

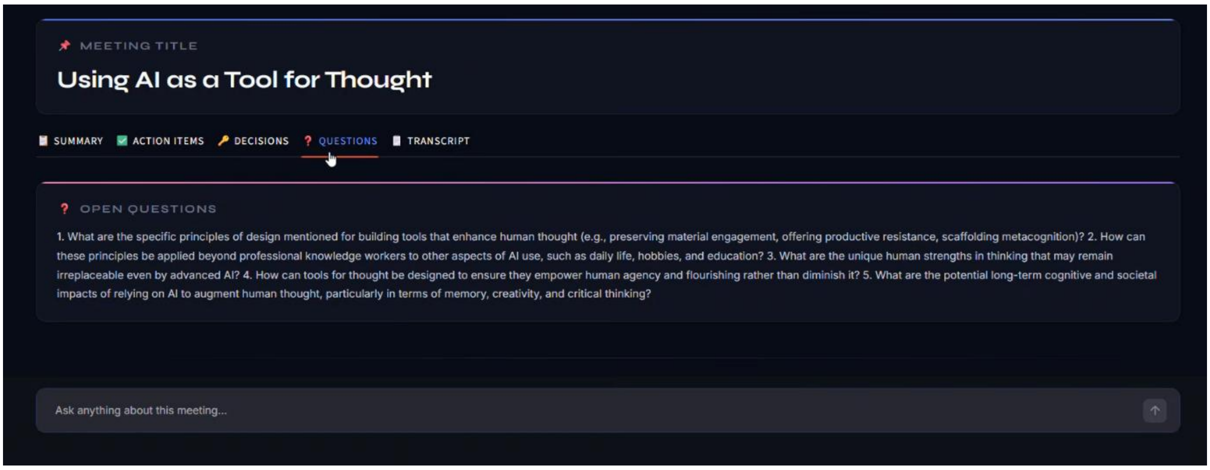
• Conversational chat interface

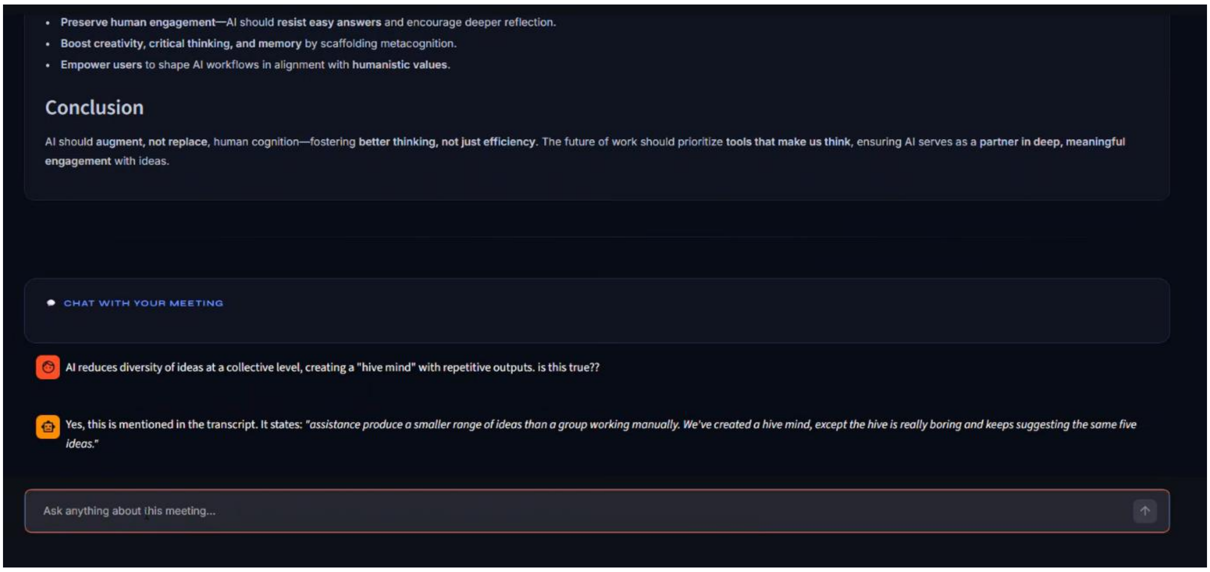
Users can upload recordings, review extracted insights, and interact with meeting knowledge through natural language questions.

The interface significantly enhances usability and user experience.









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6. TECHNOLOGIES USED

Category	Technology
Programming Language	Python
Frontend Framework	Streamlit
LLM Framework	LangChain
Large Language Model	Mistral Small Latest
Speech Recognition	OpenAI Whisper
Hinglish Transcription	Sarvam AI
Vector Database	ChromaDB
Embedding Model	all-MiniLM-L6-v2
Audio Processing	Pydub
Video Downloading	yt-dlp
Environment Management	Python-dotenv

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7. KEY FEATURES

- Automatic YouTube audio extraction.

- Local audio and video processing.
- English speech transcription using Whisper.
- Hinglish transcription using Sarvam AI.
- Automatic title generation.
- AI-powered meeting summarization.
- Action item extraction.
- Key decision identification.
- Open question detection.
- Semantic vector search.
- Retrieval-Augmented Question Answering.
- Interactive conversational interface.
- Modern Streamlit dashboard.

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8. RESULTS AND OUTCOMES

The developed system successfully converts lengthy recordings into structured knowledge representations.

Key outcomes include:

- Significant reduction in manual review time.
- Improved meeting documentation quality.
- Faster retrieval of important information.
- Enhanced accessibility of meeting content.
- Accurate context-aware conversational responses.

The system demonstrates practical application of modern Generative AI technologies in real-world productivity scenarios.

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9. FUTURE ENHANCEMENTS

Potential future improvements include:

- Speaker diarization for speaker identification.
- Support for additional regional languages.

- Sentiment and emotion analysis.
- Enterprise-scale meeting analytics dashboard.
- Cloud deployment on AWS or Azure.
- Automated email generation of meeting minutes.
- Multi-user collaboration capabilities.
- Integration with Zoom, Google Meet, and Microsoft Teams.

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10. CONCLUSION

InsightLens is a comprehensive AI-powered Video Intelligence Agent that transforms unstructured audio and video recordings into structured, searchable, and actionable knowledge.

The system integrates advanced technologies including Speech Recognition, Natural Language Processing, Large Language Models, Vector Databases, and Retrieval-Augmented Generation into a unified intelligent platform. By combining automatic transcription, summarization, meeting analytics, semantic retrieval, and conversational AI, the project goes beyond traditional video summarization and delivers a complete Meeting Intelligence Solution.

The successful implementation of InsightLens demonstrates strong capabilities in Generative AI, Information Retrieval, Conversational AI, and End-to-End AI System Development, making it a practical and scalable solution for modern knowledge management and meeting intelligence applications.